# PC5253 Final Project: Member B
# Measured Citation-Network Experiments

Member B

27 October 2026

This companion report records the five citation-network experiments assigned to B in PLAN v3.1. Counts and conclusions concern the selected AMiner database-venue induced graph and its observed time span.

## 1 Experimental design

**Citation data and time resolution.** We stream AMiner DBLP-Citation V14 (Tang et al., 2008) and retain an induced subgraph of database venues selected by a fixed whitelist (SIGMOD, VLDB/PVLDB, ICDE, PODS, EDBT, ICDT, TODS, VLDB Journal, Data & Knowledge Engineering, and Information Systems). All selected papers published in 1960–2020 are retained, including isolates. A distinct citation $u \to v$ occurs in the publication year of $u$. The primary analysis requires $y_v < y_u$; same-year citations are saved for structural sensitivity analysis because their order cannot be resolved. We exclude self-citations to the same paper, duplicate edges and future references. Author self-citations are not removed. Counts refer to citations within the selected venues, rather than global citation impact. The primary analysis years are 1990–2020, initialized using the available earlier history.

**M1: annual and cumulative citation structure.** Annual and cumulative networks use the same node denominator: all selected papers already published by year $t$. Directed density is $E/[N(N-1)]$ and mean total degree is $2E/N$. Clustering, component counts and largest-component fractions use the weak undirected projection, including isolates. Trailing windows of 1, 2, 5 and 10 years provide a window-sensitivity check; sensitivity is $(\max - \min)/\text{mean}$ over the grid at each endpoint, then averaged over endpoints.

**M2c-1: preferential attachment.** Following the degree-class approach of Jeong et al. (2003), we freeze in-degree at the start of each citing year and count eligible nodes $n_{tk}$ and new citations $h_{tk}$ in each prior-degree class $k$. We estimate $\alpha$ by conditional likelihood with probabilities proportional to $n_{tk}(k+1)^{\alpha}$, conditioning on the total number of citations in each year. The offset admits citations to previously uncited papers; offsets 0.5 and 2 provide sensitivity checks. Confidence intervals use 400 resamples of citing-year blocks. These are descriptive uncertainty intervals; neither positive $\alpha$ nor its departure from an age-only null identifies a causal mechanism.

**M2c-2: ageing.** For ages 1–60, the target probability in citing year $t$ is proportional to $n_{ta}f(a)$, where $n_{ta}$ counts available papers of age $a$. A conditional target likelihood compares discrete-grid lognormal, exponential, power-law and uniform kernels using AIC on identical observations and support. We repeat the comparison by publication cohort (1970–1989, 1990–1999, 2000–2009), accounting for each cohort's actual exposure. This estimates marginal ageing; degree, fitness and venue coverage can also influence this curve.

**M2c-3: densification and diameter.** We fit $\log E(t) = a \log N(t) + b$ over 1990–2020 and repeat over split periods and an earlier observation cutoff. Year-resampling intervals are descriptive because cumulative observations are dependent. Effective diameter is the 90th percentile of weak shortest-path distances within the largest connected component, using 128 uniformly sampled source nodes per year. Sensitivity checks use 64 and 256 sources. We save the component size and number of sources with every estimate (Leskovec et al., 2005).

**M2c-4: citation dynamics and prediction.** We fit the Wang–Song–Barabási curve (Wang et al., 2013),

$$c(t) = m\left[\exp\left\{\lambda\Phi\left(\frac{\log t - \mu}{\sigma}\right)\right\} - 1\right], \tag{1}$$

where $m$ is fixed to the mean observed within-subgraph reference count of papers published through 1989. Bounded multistart least squares minimizes squared $\log(1+c)$ errors; parameter bounds and boundary flags are preserved. For descriptive full-history fits we randomly select at most 500 papers born in 1990–2005 with at least 10 citations by age 15 and fit years 1–15. The resulting collapse is in-sample evidence only. For prediction, selection uses only the first five years: at least five observed citations by age five, followed by a fixed random sample of at most 500 papers. Using only years 1–5, we compare WSB against no-ageing PA, $c(t) = m[\exp(rt) - 1]$, and linear extrapolation, $c(5)t/5$. All models predict the same papers at age 15 and at the observed endpoint in 2020; these are censored observed totals, not unobserved lifetime citations. We report MAE, median absolute error, RMSLE, paired bootstrap differences (1000 resamples) and an interior-fit sensitivity check. Boundary fits remain in the primary evaluation so that difficult cases are not silently discarded.

**P3: citation-target randomization.** Twenty realizations per model preserve each paper's publication year and exact number of distinct observed references, with no replacement and strictly older targets. P3a selects eligible targets uniformly. P3b weights targets by a nonparametric age kernel fitted by conditional target likelihood, accounting for each citing year's available age distribution, and removes degree preference. Its age weights are calibrated from the data, so agreement on ageing is not an independent confirmation. We report null means and sample standard deviations; empirical tail probabilities have minimum resolution 1/21. Both models fix $N(t)$ and $E(t)$ and hence exactly preserve the densification exponent. They can test diameter and target preference, but cannot supply a significance test for densification itself.

**Reproducibility.** All significant CPU workloads run on Vanda compute nodes through PBS without GPU allocation. Fixed seeds, input/source checksums, environment lock file, filtering counts, individual forecasts and every randomized edge list are saved. Robustness checks cover time splits, cutoff 2015, degree offsets, same-year edges, core database venues, diameter sampling, and 10 versus 20 null realizations.

# 2 Measured results

## 2.1 Data and preprocessing audit

The official V14 archive[1] contains 5,259,858 paper records. The fixed database-venue rules match 22,237 records; excluding 3,087 publications outside 1960–2020 leaves 19,150 papers spanning 1970–2020. Venue families include related workshops and conference special issues matched by the declared patterns. We retain isolates; 6,709 papers have no incident edge in the primary induced graph. The 263,648 raw reference entries of selected source papers are partitioned in Table 1. The retained 73,269 citations are distinct and point to strictly older selected papers. Recorded-year inconsistencies may reflect metadata or preprint/publication-date differences, so their removal is a

[1] https://originalfileserver.aminer.cn/misc/dblp_v14.tar.gz

convention, not a claim that every excluded citation is erroneous. The archive is distributed by AMiner for research use; no additional redistribution licence is inferred.

| Reference-entry category | Count |
|---|---|
| Target outside the selected subgraph | 140,569 |
| Repeated reference in the same source record | 37,440 |
| Source and target are the same paper | 494 |
| Target has a later recorded publication year | 6,390 |
| Same-year reference, saved separately | 5,486 |
| Primary strictly older-target citation | 73,269 |
| Total selected-source reference entries | 263,648 |

Table 1: Exact reference-entry audit. Categories sum to the original entry count.

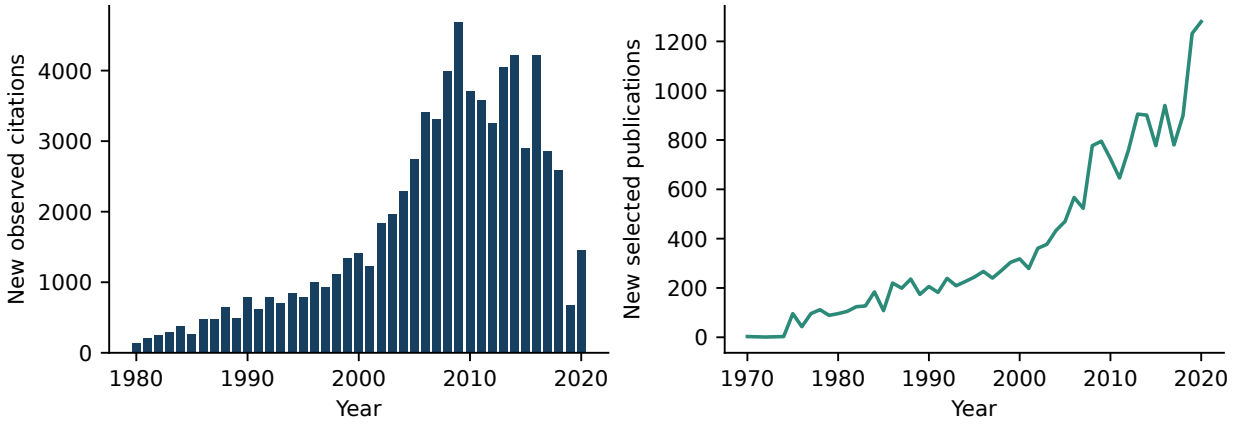



Figure 1: Selected publications and observed within-subgraph citations per year. The fall in late-period references, despite growing publication counts, cautions against interpreting the boundary as a change in scientific behaviour.

## 2.2 M1: annual versus cumulative structure (H1)

On the common set of 19,150 born nodes in 2020, the annual graph contains 1,450 citations and the cumulative graph 73,269. Mean total degree is respectively 0.151 and 7.652; directed density is $3.95 \times 10^{-6}$ and $2.00 \times 10^{-4}$. The weak largest-component fractions are 0.0523 and 0.6405, with 17,966 and 6,782 weak components. Cumulative weak clustering is 0.1241. Annual weak clustering is exactly zero in every year: each retained annual edge joins a new paper to an older paper, making that snapshot bipartite. This zero is therefore a consequence of the year convention, not independent evidence for a social or temporal mechanism.

Over 1-, 2-, 5- and 10-year trailing windows, mean relative ranges are 1.798 for both density and mean degree, 2.361 for weak clustering, and 1.114 for the largest-component fraction. Density and degree have identical sensitivity here because the node denominator is fixed within an endpoint. **Verdict on H1:** structural descriptions depend strongly on window choice in this induced citation graph; some differences are definitional rather than independent discoveries about temporal correlations.

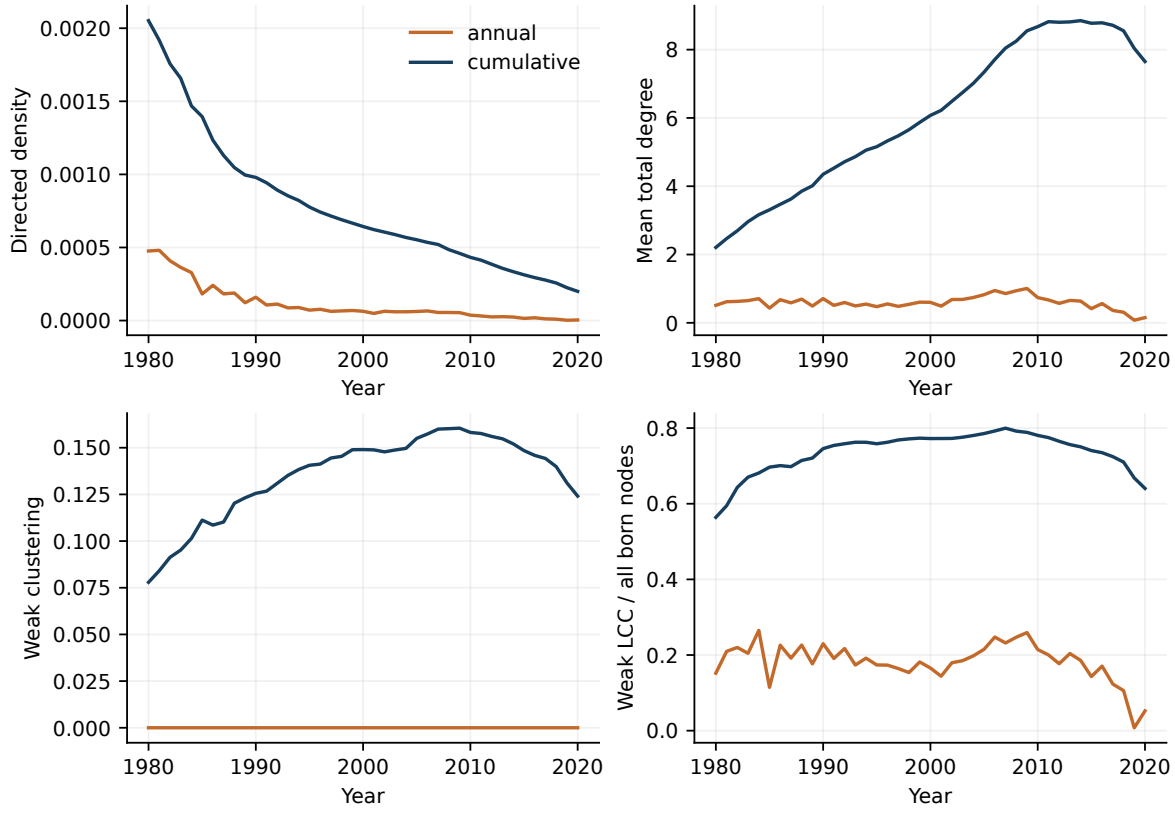



Figure 2: Measured annual and cumulative graphs with a common born-node denominator. Clustering and connectivity use the weak projection. Zero annual clustering is imposed by strict publication-year ordering.

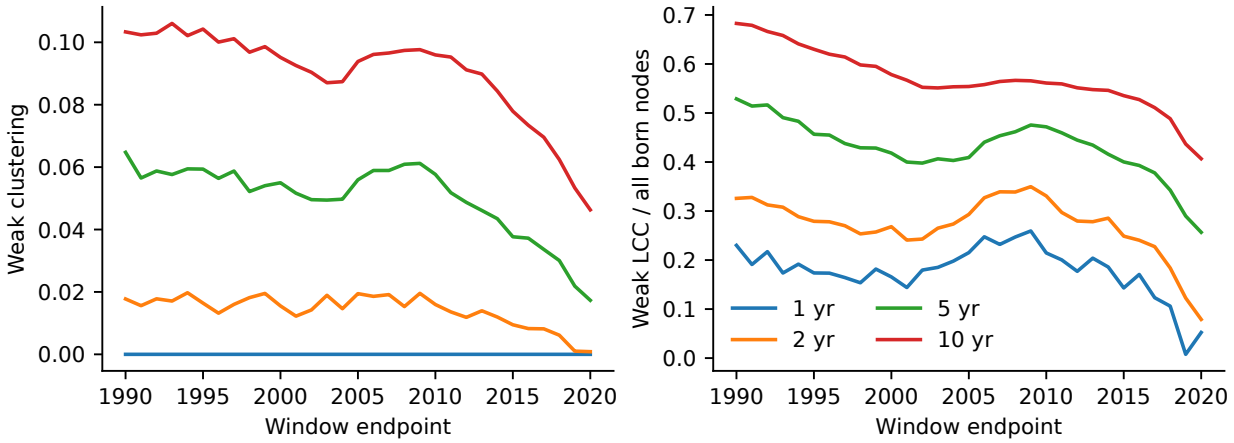



Figure 3: Clustering and connectivity under trailing windows. All panels include already-published isolates; changing the denominator would answer a different question.

### 2.3 M2c-1: preferential attachment (H4a)

The conditional degree kernel yields $\alpha = 0.7136$, with descriptive 95% citing-year bootstrap interval $[0.6928, 0.7297]$ for $\Pi(k) \propto (k+1)^{\alpha}$. Across 20 realizations, P3a gives $0.00059 \pm 0.00508$ and the calibrated age-only P3b gives $-0.65118 \pm 0.00532$ (mean $\pm$ sample SD). The observed exponent exceeds every null value; the one-sided empirical probability is $1/21 = 0.0476$ for each comparison. This is the coarse minimum resolution of 20 draws, not a high-precision significance claim or a multiple-testing-adjusted result.

The positive association persists for degree offsets 0.5 and 2 ($\alpha = 0.6040, 0.8576$), the two observation halves (0.6844, 0.7214), cutoff 2015 (0.7218), and the core SIGMOD/VLDB/ICDE venue-family subset (0.7309). The numerical exponent depends on the offset. **Verdict on H4a:** evidence is consistent with sublinear degree preference beyond an age-only reference model. Unobserved paper fitness, venue coverage and other heterogeneity prevent a causal interpretation of $\alpha$.

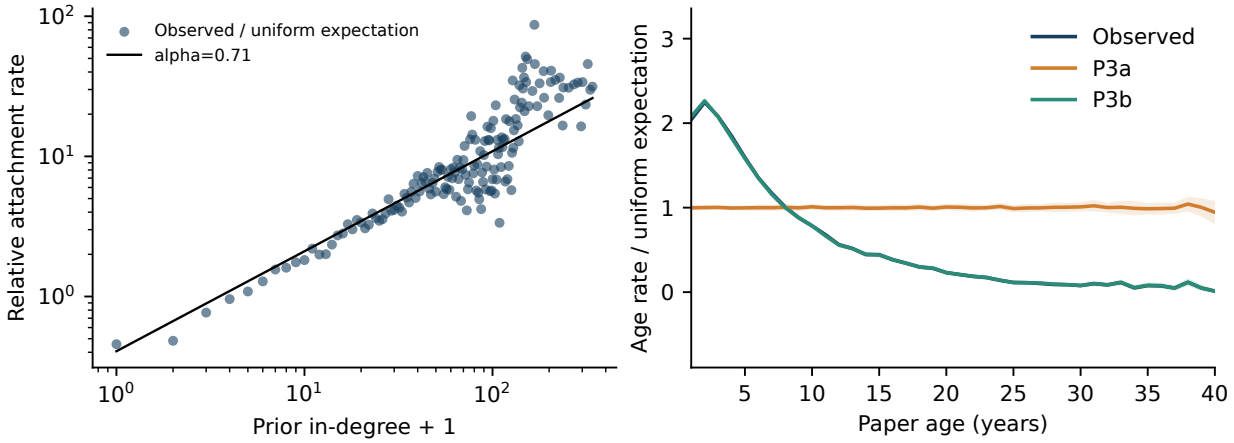



Figure 4: Prior-degree attachment and exposure-adjusted ageing. Kernel points with at least five hits are shown for legibility; the conditional exponent uses all supported degree classes. The display line uses a pooled normalization, whereas the estimator conditions on each citing year. Null bands are mean ± SD. P3b's ageing agreement is expected from its in-sample calibration.

## 2.4 M2c-2: ageing and model comparison (H4b)

Among the declared families, the lognormal target-preference kernel has the lowest AIC (Table 2). Its parameters are $\mu = 1.8584$ and $\sigma = 1.1078$, corresponding to the underlying continuous lognormal scale $\exp(\mu) = 6.41$ years. This scale is not the mean age of the observed citations, because annual target availability also enters the likelihood. Across all retained events, mean citation age is 5.711 years, versus $11.457 \pm 0.034$ under P3a and $5.717 \pm 0.018$ under P3b. The latter similarity validates the fitted control's calibration rather than supplying independent evidence for ageing.

| Age kernel | Number of parameters | ΔAIC |
|---|---|---|
| Lognormal | 2 | 0.0 |
| Exponential | 1 | 537.8 |
| Power-law decay | 1 | 9,360.5 |
| Uniform target choice | 0 | 41,796.7 |

Table 2: Models compared on the same observations and at-risk age support. The grid permits ages up to 60; this dataset supplies at most 50 years of exposure.

Lognormal is also preferred within all three target-publication cohorts (1970–1989, 1990–1999 and 2000–2009), and in both observation halves. However, fitted cohort parameters differ: for 2000–2009 papers the estimated $\exp(\mu)$ is about 25.5 years, beyond that cohort's longest available follow-up. Its unobserved tail must not be interpreted as a measured lifetime. **Verdict on H4b:** age-independent target choice is inadequate in these data; a lognormal age-preference kernel is best among the tested forms. The result is marginal ageing and does not separately identify obsolescence, fitness or coverage.

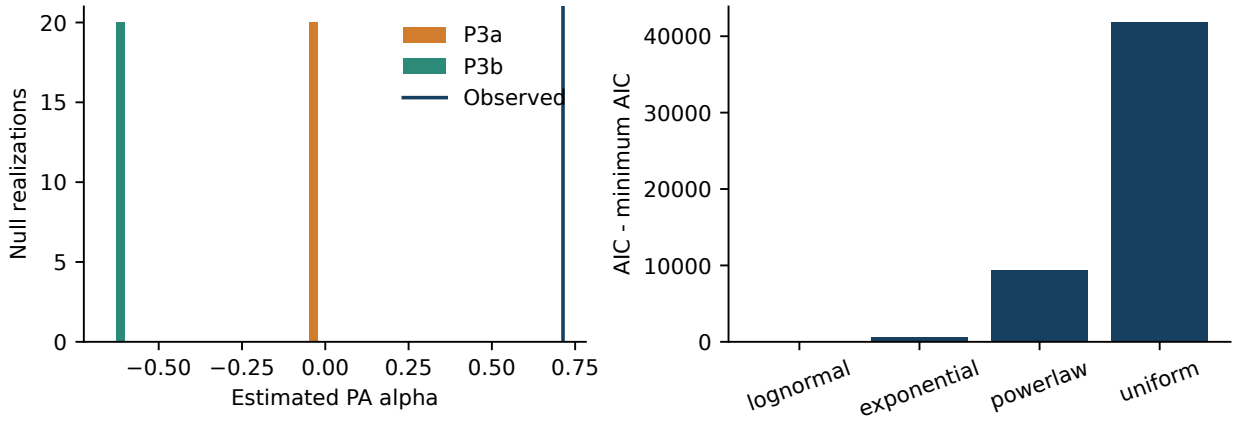



Figure 5: Degree exponents under P3a/P3b and relative AIC of age kernels. These diagnostics condition on the selected corpus and the declared risk sets.

### 2.5 M2c-3: densification without diameter shrinkage (H4c)

The cumulative graph follows a descriptive densification relation with $a = 1.3426$, $R^2 = 0.9913$, and a year-resampling interval $[1.2876, 1.4062]$. Densification means superlinear edge growth relative to nodes, not an increase in $E/[N(N-1)]$: directed density actually falls while cumulative mean degree rises from 4.350 in 1990 to 7.652 in 2020. The global exponent masks weaker late growth: $a = 1.4548$ in 1990–2004 and 1.0646 in 2005–2020. It is 1.4270 with cutoff 2015, 1.3242 when the 5,486 same-year references are included, and 1.4099 in the core venue-family subset ($N = 14,235$, $E = 47,009$).

The sampled weak-LCC effective diameter is 6 hops in every year from 1990 to 2020. The largest component grows from 1,657 to 12,266 nodes. Repeating the 1990, 2005 and 2020 estimates with 64 and 256 sources again gives 6 hops. In 2020 the null estimates are $5.40 \pm 0.50$ (P3a) and $5.95 \pm 0.22$ (P3b). P3 leaves $N(t)$ and $E(t)$ unchanged, so it exactly preserves $a$ and supplies no significance test for densification. **Verdict on H4c:** superlinear cumulative growth is observed, but diameter shrinkage is not detected at the sampled 90th-percentile resolution. These are separate empirical propositions.

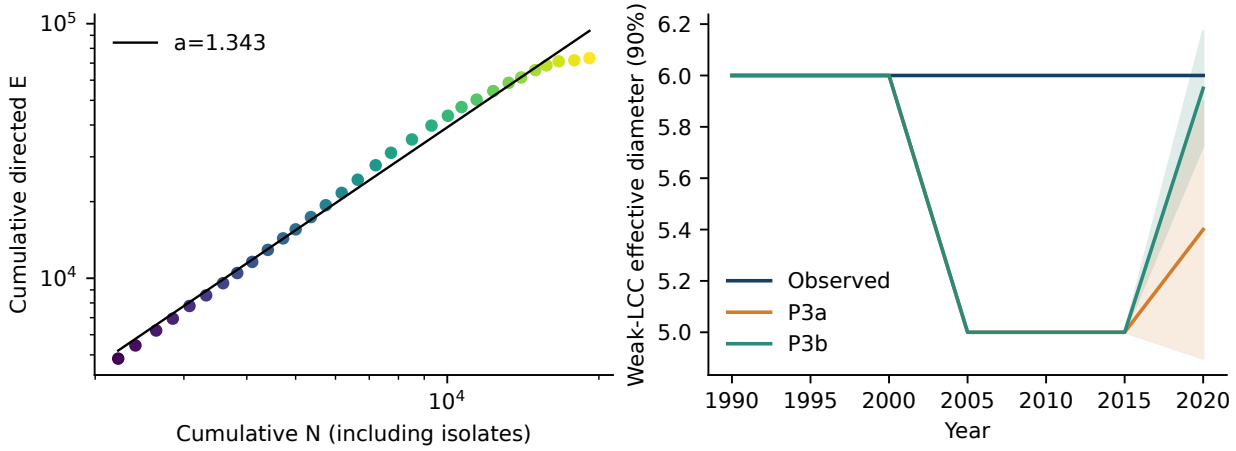



Figure 6: Densification and sampled weak-LCC diameter. Colour in the left panel runs from early (dark) to late (yellow) years. Null means and SD are evaluated every five years. A stable integer-valued 90th percentile does not imply that every part of the distance distribution is unchanged.

### 2.6 M2c-4: compact fits versus predictive evidence (H4d)

The historical reference parameter is fixed at $m = 2.00645$. Among 1,019 eligible papers with at least ten citations by age 15, the prespecified random sample of 500 full histories has median parameters $(\lambda, \mu, \sigma) = (2.7167, 0.8343, 1.2375)$. Their interquartile ranges are respectively $[2.2151, 3.4305]$, $[0.4075, 1.3523]$ and $[0.9459, 1.7245]$. Median full-history RMSLE is 0.0736, and median scaled-collapse RMSE is 0.0220 (90th percentile 0.0453). Full-history boundary fits occur for 10.2% of papers. These small fitted-sample errors describe calibration; they do not independently prove a universal curve.

Forecast selection uses only the first five years, yielding 1,220 eligible papers and a fixed random sample of 500. Every selected fit converges, but 38.6% hit a parameter bound, indicating weak identification from short histories. All 500 remain in the primary comparison (Table 3). The no-ageing PA baseline can extrapolate explosively; beating it alone is a weak predictive test.

| Target | Model | MAE | Median AE | RMSLE |
|---|---|---|---|---|
| Age 15 | WSB | 15.144 | 5.433 | 0.601 |
| | Linear | 17.064 | 13.000 | 0.662 |
| | PA, no ageing | 27,588.0 | 945.937 | 4.601 |
| Observed 2020 | WSB | 28.016 | 6.786 | 0.778 |
| | Linear | 27.271 | 20.900 | 0.876 |
| | PA, no ageing | $2.79 \times 10^6$ | 6,466.682 | 7.000 |

Table 3: Matched predictions using only ages 1–5 for training, $n = 500$ in every row. The 2020 horizon varies with birth year and remains an observed, censored within-corpus total, not a lifetime citation count.

The paired bootstrap MAE difference (WSB minus linear) is $-1.920$ with 95% interval from $-4.471$ to 0.806 at age 15, and 0.745 with an interval from $-4.764$ to 8.128 at the 2020 endpoint. Both include zero. WSB has smaller RMSLE and median error, but no clear mean-absolute-error advantage over linear extrapolation. Restricting to the 307 interior fits improves WSB's age-15 MAE to 9.991, versus 18.547 for linear, but this selected subset cannot replace the all-paper comparison. **Verdict on H4d:** three parameters compactly describe these selected histories; superiority over a simple linear predictor is not established.

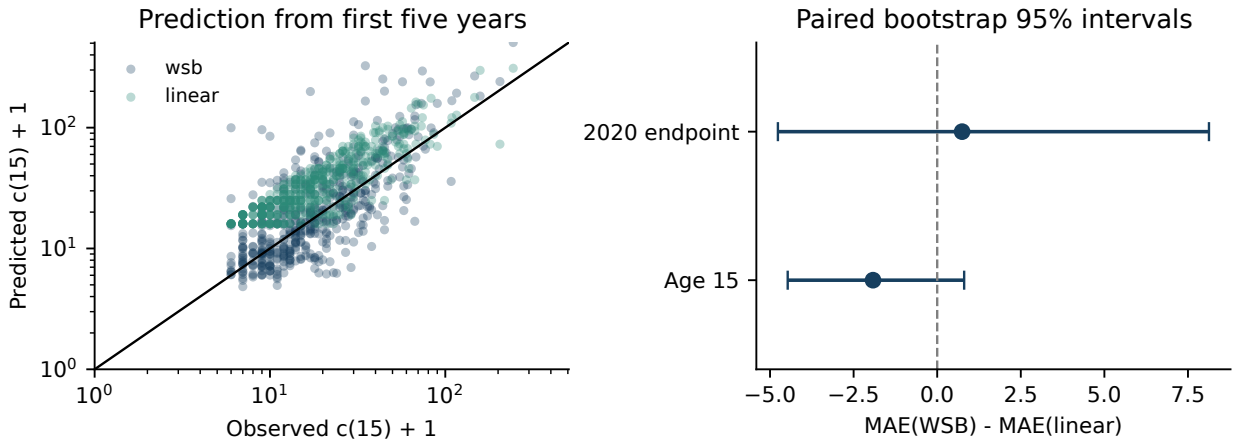



Figure 7: Held-out prediction and the paired MAE differences versus linear. Negative differences favour WSB; both 95% intervals cross zero. The much weaker no-ageing PA baseline is reported in Table 3 and the full three-panel diagnostic asset `b_wsb.pdf`.

### 2.7 Reproducibility, robustness and scope

All five experiment blocks and 40 null graphs completed in serial PBS main job 1500361 on CN-016, with four requested CPU cores, no GPU, exit status 0 and 2 min 54 s wall time including the archive stage. Preparation job 1500299 also exited 0. Thirteen known-answer unit tests and the formal raw-edge/forecast selection audit passed. Every null preserves exact source/year counts, distinct references and strictly older targets. First-ten versus all-twenty mean exponents are 0.00055 versus 0.00059 for P3a and −0.65320 versus −0.65118 for P3b.

The exact environment lock, seeds, individual forecasts, fitted parameters, bootstrap draws, all null edge lists, original-year ties and filtering counts are saved in the shared summary table (`results/b_summary.csv`). The raw archive SHA-256 is

`729cc2c107d8abbe586d9801340246fd94e94d916ba1cebe64dd93a399f0cd28`

and matches its persistent HPC-home copy. Conclusions concern one induced subfield corpus. Missing external citations, venue-name coverage, recorded-year inconsistencies, annual time resolution, changing late-period reference coverage, right censoring and fitted-sample selection limit generalisation. No comparison with PS or CM is inferred before their measured results are available.

## References


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